Topics included in this document are as follows:

- JobBOSS XML SDK Overview
- Prerequisites and Assumptions
- Documentation Conventions
- Supported Functions
- Communicating with JobBOSS Using XML
- JobBOSS XML Document Requirements
- JBXMLRespond
- JBXMLRequest

JobBOSS XML SDK Overview

The JobBOSS XML SDK provides a means to use XML formatted data with the JobBOSS database. The SDK implements a method that processes XML documents sent to JobBOSS, allowing you to maintain data (add, modify, and delete records) and query for both individual records and record lists. For example, you can add a collection of customer records, or query to find a list of jobs for a given company name.

XML (Extensible Markup Language) provides a standardized way to describe data exchanged between software applications. Using XML, you can describe almost any data, in a format that is understood by the majority of computers in use today.

These instructions assume that you, as a developer communicating with the JobBOSS database using XML, are already familiar with XML, its format, and its structure. If not, we highly recommend that you acquire some education on the topic, either independently or formally through a class.

Prerequisites and Assumptions

We assume the programmer using this documentation is familiar with basic XML concepts and usage.

The JobBOSS XML SDK uses the Microsoft XML Core Services (MSXML) 4.0, which is installed with the rest of the JobBOSS XML SDK. You can either use the MSXML parser, or another parser, when creating your XML files.

The JobBOSS XML SDK supports the following databases:

- Microsoft Access
- Microsoft SQL Server

In order to deploy the JobBOSS XML SDK, you need to have the JobBOSS software installed on your workstation. JobBOSS XML SDK will be automatically installed along with rest of the JobBOSS client software.

Note: If you have more than one JobBOSS database (that is, you have more than one database alias defined in JobBOSS), the JobBOSS XML SDK will operate against the database you are currently logged into, or the most recent one you were logged into on that workstation.

You need to list elements in your XML documents in the order specified, as there may be later dependencies upon a given functional request.

Documentation Conventions

Conventions used in this document are as follows:

- Elements can be either required or optional and are listed in their required sequence.
- Complex elements (those with child elements) are each portrayed as a hyperlink. Further information about child elements (or attributes) is accessible by clicking each link.
- Simple elements (those without child elements) list their data type after the element name.
- Most often, each element can appear only once in the sequence; those which can appear more than once are noted as such.
- Predefined values are listed for elements that have them. Attributes which have values that are predefined by the user are noted as such.

Supported Functions

Business Data Maintenance Functions

JobBOSS XML SDK supports the functions marked in this table:

Function	*Add*	*Modify*	*Delete*	*Query*	*List Query*
Customer	X	X	X	X	X
Vendor	X	X	X	X	X
Service Operation	X	X	X	X	X
Work Center	X	X	X	X	X
Material	X	X	X	X	X
Job	X	X	X	X	X
Department					X
Reference					
Customer Type					X
Tax Code					X
Ship Via					X
Alloy					X
Material Class					X
Discount Level					X
Terms					X
Currency					X
Sales Code					X
Location Type					X
Warehouse					X
Additional Cost					X
Standard Text					X
Time Entry	X	X	X	X	X
Sales Order	X	X	X	X	X
Quote	X	X	X	X	X

Here are some example requests:

- JobQueryRq-shows everything about a job.
- CustomerQueryRq-shows everything about a customer.
- JobAddRq-adds a new job.
- CustomerAddRq-adds a new customer.

You can use any ListQueryRq to find data to use elsewhere in other requests (element names correspond to fields in the JobBOSS database).

For queries that could potentially return large amounts of data, be sure to use a filter. For example, if you were to request all jobs with no filter, you might experience performance degradation.

Wildcard Support

For any Query or ListQuery function, you can use the% (percent) symbol as a wildcard character in a query element. For example, this request returns all customers whose identifier begins with "CO":

```
<CustomerListQueryRq>
  <QueryFilter>
    <IDFilter>C0%</IDFilter>
  </QueryFilter>
</CustomerListQueryRq>
```

Sample XML, XSD, and Executable Files

The following files are installed during the JobBOSS Client installation, in the JobBOSS folder:

File Name	*Description*
Request.xml	sample request file
Respond.xml	sample response file
Errors.xml	sample response file with errors
JBXML.xsd	schema file
JBXMLMap.xml	a mapping file to JobBOSS objects
Jbinterface.exe	a Delphi executable file

User-Defined Values

User-defined values are specified in JobBOSS's System Administration module, through the Preferences function, under the User Codes tab. If you try to use the JobBOSS XML SDK to specify a user-defined value which has not been defined in the JobBOSS database as a user code, you will get an error. For example, if you try to add a customer (CustomerAddRq) using a value for TennsCodeRef that has not been defined as a valid terms user code, you will get an error.

On the other hand, if a terms code was valid at the time a record was added, but later this terms code was deleted from the user codes, and you try to modify a customer (CustomerModRq) that uses this term's code, no error will result.

Communicating with JobBOSS Using XML

Creating the COM Object

To communicate with JobBOSS using XML, you need to use the JobBOSS COM object provided with the SDK. To use the COM object, you need to write some code (using a language such as Delphi, Visual Basic, or some other language) to create the JobBOSS COM object for you. The COM object instance provides you with the interface required to communicate with JobBOSS.

To create and connect to the JobBOSS COM object, you will need to do the following:

1. Reference the JobBOSS COM object in your application development project. This object (provided by Jbinterface.exe) is called JBRequestProcessor.
2. Create the JBRequestProcessor object in your code.
3. Communicate with the JBRequestProcessor object using its ProcessRequest method. The signature of this method is as fo11ows:
   - Takes an XML document (a string, for an incoming XML request file)
   - Returns an XML document (a string, for an outgoing XML response file)

Licensing Requirements

When you use the SDK, it will take up a JobBOSS seat license by requiring a session ID for all XML requests. To obtain a session ID, you need to call CreateSession using a valid JobBOSS user name and password. This returns a Session ID which you can then use for all requests during that session. When all requests have been completed for a session, you need to call CloseSession and pass the corresponding session ID to release the seat license. Any requests that do not have a valid session ID will return an error.

How You Might Use the SOK

Following are some examples of how you might use the SDK:

- You could insert existing data stored in an Excel spreadsheet into your JobBOSS database.
- You could write an input fonn in Visual Basic that accepts your data, submits it to JobBOSS to validate according to its business rules, and then insert this data into your JobBOSS database.
- You could use an Excel spreadsheet as your input form and to provide user defined validation to your data. This allows you to specify business rules for the Excel form to enforce *prior to* inserting the data into your JobBOSS database, which then enforces its own validation rules. (To do this, use Visual Basic for Applications to map ce11s in the Excel spreadsheet to the data with Visual Basic procedures.)

Sample Visual Basic Code

Following is one example of a Visual Basic code snippet that uses the JobBOSS COM object to process functional requests against the JobBOSS database (this example happens to use MSXML 4.0, although you are not required to do so):

```
Public Sub ProcessXMLRequests()

    ' Purpose: Take an XML document containing JobBOSS functional
    ' requests, process its transactions against the JobBOSS database,
    ' and save the resulting response data in an XML file.

    ' Before calling this code, you need to set a reference to
    'Microsoft XML, v4.0 in your project.

    ' Set up space for an object variable for the JobBOSS COM object.
    Dim jb As JBRequestProcessor

    ' Set up space for object variables used to contain request and
    ' response strings.
    Dim Request,Response As String

    ' Set up space for object variables for MSXML DOMDocument objects.
    ' These object variables represent XML documents that hold response
```

```
    ' and request data, respectively.
    Dim ResDoc,ReqDoc As Msxml2.DOMDocument40
    Dim SessionID As String

    ' Create the JobBOSS COM object.
    Set jb = New JBRequestProcessor
    SessionID = jb.CreateSession(UserName, Psw, ErrorMsg)

    ' Create the MSXML DOMDocument objects for a request and response.
    Set ReqDoc   New Msxml2.DOMDocument40
    Set ResDoc = New Msxml2.DOMDocument40

    ' Load request data from a hard-coded file path to an XML file.
    ReqDoc.async = False
    ReqDoc.Load ("C:\MyXML\MyRequestData.xml")

    ' Create a string from the data in the ReqDoc object variable and
    ' store it in the Request object variable. In particular, the xml
    'method is called on the ReqDoc object variable, and the resulting
    ' string is stored in a Request string.
    Request=  ReqDoc.xml

    ' Pass the Request string to the jb object, which processes the
    ' JobBOSS database transactions as specified in the functional
    ' requests contained in this string of XML data, and return a
    ' Response string (in XML format).
    Response=  jb.ProcessRequest(Request)

    ' Load the XML response data into the ResDoc object variable. The
    ' LoadXML method (from the MSXML DOM document object) takes as its
    ' argument a Response string.
    ResDoc.LoadXML(Response)

    ' Save response data to a hard-coded file path to an XML file.
    ResDoc.save ("C:\MyXML\MyResponseData.xml")
    Jb.CloseSession(SessionID)

End Sub
```

JobBOSS XML Document Requirements

Every XML document used to communicate with the JobBOSS interface must contain the root element JBXML, used to format information according to an XML schema. In addition, the file will contain *one* of the following elements:

- JBXMLRequest—each document *sent to* JobBOSS will contain one JBXMLRequest element, which can contain multiple functional requests.
- JBXMLRespond—each document *received from* JobBOSS will contain one JBXMLRespond element, which can contain multiple functional responses (one for each corresponding request).

This document describes all the functional *request* elements used to send data to JobBOSS, plus selected functional *response* elements that return data from JobBOSS. To find out about the rest of the response elements, refer to the sample Respond.xml file in your installation.

CAUTION: In order for your XML document to work properly, you need to enter elements *in the order listed.* The sequence specified is *required* to match the JobBOSS XML document schema. If elements are entered in a different sequence, JobBOSS will *not* process your data.

JBXMLRespond

Upon completing JBXML requests, JobBOSS generates a response file in XML format. The response file includes a root JBXML element and a JBXMLRespond element, plus one or more individual functional response elements. This file sends back the data you supplied in your JBXML request as well as any default data supplied by JobBOSS.

Response Data

The response file provides information which you may need to know in order to create other requests. For example:

- A LastUpdated element provides the date and time when something was last changed, which can be useful to know when multiple users are maintaining the same data. This element is available for customer add and modification requests (CustomerAddRq and CustomerModRq), among others.
- Numerous calculated fields are available, such as EstimatedTota!Hours, EstimatedTotalTime, ActualTotalTime, RemainingTotalTime, EstimatedTotalCost, and ActualTotalCost.
- Numeric identifiers (IDs) are sent back for automatically numbered fields.

Error Handling and Reporting

These error codes report the status of an XML request; for each functional request that generates an error, its description is listed in the XML response file:

The description of the error is available in the XML response file.

Code	*Description*
0 (zero)	success
1	business logic problem (e.g., no customer, customer is over their credit limit, vendor is not in the JobBOSS database, or record already exists)
2	COM object failed

If errors occur, the entire functional request transaction is not committed. For example, say you have an XML request file that includes three customer add requests (that is, three CustomerAddRq elements). If one CustomerAddRq element has errors and the other two don't, the one with errors will not be added, but the other two will be. Refer to Business Data Maintenance Functions for a list of all functional requests.

Type Formatting

The JobBOSS XML SDK uses ISO standard date and date-time formats, as well as string, boolean, date, int, double, and float types.

The ISO standard date format is: YYYY-MM-DDThh:mrn:ss

Representation	*Description*
YYYY	Four-digit year
MM	Two-digit month (01 through 12). January would be 01
DD	Two-digit day of month (01 through 31)
hh	Two digits of hour (00 through 23). Note the am or pm designation is not allowed
mm	Two digits of minute (00 through 59)

ss	Two digits of second (00 through 59)

All the components in the date format must be present for a field of type Date, with the punctuation exactly as shown. If the field is designated as a Date/Time field, then the components of the time format must also be present. Note that the" T" appears *literally* in the string to indicate the beginning of the time element and must be uppercase. You may use "00" for the seconds, minutes or hours when the exact time is unknown.

XML requires special characters to be specified as shown:

To insert:	*Which is called:*	*Use this entity reference:*
&	ampersand	&
	apostrophe (single quote)	'
"	quotation mark (double quote)	"
>	greater-than symbol	>
<	less-than symbol	<

For example:

XML code:	*Result:*
<Name>Butch & Sundance</Name>	Butch & Sundance
<Name>Sean O'Neil</Name>	Sean O'Neil
<Name>Larry "Bud" Melman</Name>	Larry "Bud" Melman
<Formula>(x > y) AND (x &It; z)</Formula>	(x > y) AND (x < z)

Schema Information and XML Usage

The XML schema document (JBXML.xsd) included with your installation specifies the names of elements, the sequence in which they are used, whether they are required or optional, and their types.

Following are some XML usage considerations to keep in mind:

- Use well-formatted XML, marked with open and close tags (<tagname> and </tagname>).
- You can validate your XML request against the schema using an XML editor to check for parser errors; this will validate the following:
  - Information is in the correct order
  - Tags have the correct names (names are case-sensitive)
  - Date-time format is correct

You need to know length limits for your data elements. The size of element data is the same as the size of the corresponding fields in your JobBOSS database. You can automatically generate this information about your JobBOSS database using the JobBOSS Database Schema program (c:\Program Files\JobBOSS\DBSchema.exe).

JBXMLRequest

JobBOSS requires an ID and LastUpdated element for any modify request. The method used to obtain the correct LastUpdated date/time is to first send the associated query request. For example, if you wanted to do a MaterialModRq, you would need to send a MaterialQueryRq first to retrieve the LastUpdated value to use in the modify request. The LastUpdated date/time stamp ensures that only one update can be made to a record at one time.

This document lists all possible XML tags which can be used to send data to JobBOSS. Typically, you would need to use only a small subset of these elements at a time.

- CustomerAddRq Can have multiple.
- CustomerModRq Can have multiple.
- CustomerDelRq Can have multiple.
- CustomerQueryRq Can have multiple.
- CustomerListQuery_Rq_ Can have multiple.
- VendorAddRq_ Can have multiple.
- VendorModRq_ Can have multiple.
- VendorDelRq Can have multiple.
- VendorQuery_Rq_ Can have multiple.
- VendorListQueryRq_ Can have multiple.
- ServiceOperationAddRq_ Can have multiple.
- ServiceOperationModRq_ Can have multiple.
- ServiceOperationDelRq_ Can have multiple.
- ServiceOperationQueryRq_ Can have multiple.
- ServiceOperationListQuery_Rq_ Can have multiple.
- WorkCenterAddRq_ Can have multiple.
- WorkCenterModRq_ Can have multiple.
- WorkCenterDelRq_ Can have multiple.
- WorkCenterQueryRq_ Can have multiple.
- WorkCenterListQuery_Rq_ Can have multiple.
- StockItemAddRq Can have multiple.
- StockItemModRq Can have multiple.
- StockItemDelRq Can have multiple.
- StockItemQueryRq Can have multiple.
- StockItemListQuery_Rq Can have multiple.
- MaterialAddRq Can have multiple.
- MaterialModRq_ Can have multiple.
- MaterialDelRq_ Can have multiple.
- MaterialQueryRq_ Can have multiple.
- MaterialListQuery_Rq_ Can have multiple.
- JobAddRq_ Can have multiple.
- JobModRq_ Can have multiple.
- JobDelRq_ Can have multiple.
- JobQueryRq_ Can have multiple.
- JobListQueryRq_ Can have multiple.
- LocationListQueryRq_ Can have multiple.
- DepartmentRefListQueryRq_ Can have multiple.
- CustomerTypeListQueryRq_ Can have multiple.
- TaxCodeListQueryRq_ Can have multiple.
- ShipViaListQueryRq_ Can have multiple.
- Alloy_ListQuery_Rq_ Can have multiple.
- MaterialClassListQueryRq_ Can have multiple.
- DiscountLevelListQueryRq_ Can have multiple.
- TermsListQueryRq_ Can have multiple.
- CurrencyListQueryRq_ Can have multiple.
- SalesCodeListQueryRq_ Can have multiple.

- LocationTypeListQuery Rq Can have multiple.
- WarehouseListQueryRq Can have multiple.
- AdditionalCostListQueryRq Can have multiple.
- StandardTextListQueryRq Can have multiple.
- MaterialConversionQueryRq Can have multiple.
- TimeEntryListQueryRq Can have multiple.
- SalesOrderAddRq Can have multiple.
- SalesOrderModRq Can have multiple.
- SalesOrderDelRq Can have multiple.
- SalesOrderQueryRq Can have multiple.
- SalesOrderListQueryRq Can have multiple.
- QuoteAddRq Can have multiple.
- QuoteModRq Can have multiple.
- QuoteDelRq Can have multiple.
- QuoteQuery Rq Can have multiple.
- QuoteListQueryRq Can have multiple.

CustomerAddRq

Required Elements

- CustomerAdd

Optional Elements

- AddressAdd Can have multiple.
- ContactAdd Can have multiple.
- StartUpDocumentAdd Can have multiple.
- SetAddressDefault Can have 3.

CustomerModRq

Required Elements

- CustomerMod

Optional Elements

- AddressDel Can have multiple.
- AddressMod Can have multiple.
- AddressAdd Can have multiple.
- ContactDel Can have multiple.
- ContactMod Can have multiple.
- ContactAdd Can have multiple.
- StartUpDocumentAdd Can have multiple.
- StartUpDocumentDel Can have multiple.
- StartUpDocumentMod Can have multiple.
- SetAddressDefault Can have 3.

CustomerDelRq

Required Elements

- CustomerDel Can have multiple.

CustomerQueryRq

Required Elements

- CustomerQueryFilter

CustomerlistQueryRq

Required Elements

- QueryFilter

VendorAddRq

Required Elements

- VendorAdd

Optional Elements

- AddressAdd Can have multiple.
- ContactAdd Can have multiple.
- ServiceContractAdd Can have multiple.
- VendorStartUpDocAdd Can have multiple.
- SetAddressDefault Can have 3.

VendorModRq

Required Elements

- VendorMod

Optional Elements

- AddressDel Can have multiple.
- AddressMod Can have multiple.
- AddressAdd Can have multiple.
- ContactDel Can have multiple.
- ContactMod Can have multiple.
- ContactAdd Can have multiple.
- StartUpDocumentDel Can have multiple.
- VendorStartUpDocMod Can have multiple.
- VendorStartUpDocAdd Can have multiple.
- ServiceContractDel Can have multiple.
- SeviceContractMod Can have multiple.
- ServiceContractAdd Can have multiple.
- SetAddressDefault Can have 3.

VendorDelRq

Required Elements

- VendorDel

VendorQueryRq

Required Elements

- VendorQueryFilter

VendorlistQueryRq

Required Elements

- QueryFilter

ServiceOperationAddRq

Required Elements

- ServiceOperationAdd

Optional Elements

- ServiceContractAdd Can have multiple.

ServiceOperationModRq

Required Elements

- ServiceOperationMod

Optional Elements

- ServiceContractDel Can have multiple.
- SeviceContractMod Can have multiple.
- ServiceContractAdd Can have multiple.

ServiceOperationDelRq

Required Elements

- ServiceOperationDel

ServiceOperationQueryRq

Required Elements

- ServiceOperationQueryFilter

ServiceOperationlistQueryRq

Required Elements

- QueryFilter

WorkCenterAddRq

Required Elements

- WorkCenterAdd

WorkCenterModRq

Required Elements

- WorkCenterMod

WorkCenterDelRq

Required Elements

- WorkCenterDel

WorkCenterQueryRq

Required Elements

- WorkCenterQueryFilter

WorkCenterlistQueryRq

Required Elements

- QueryFilter

StockItemAddRq

Note: Stock Item records are used to group **Raw Stock** materials together.

Required Elements

- StockItemAdd
- SetShape
- EditAlloy

Optional Elements

- EditDimension Can have multiple.
- WeightFactorMod
- VendorPartAdd Can have multiple.

StockItemModRq

Required Elements

- StockItemMod

Optional Elements

- SetShape
- EditAlloy
- EditDimension Can have multiple.

- WeightFactorMod
- VendorPartMod — Can have multiple.
- VendorPartDelete — Can have multiple.
- VendorPartAdd — Can have multiple.

StockItemDelRq

Required Elements

- StockItemDel

StockItemQueryRq

Required Elements

- StockItemQueryFilter

StockItemlistQueryRq

Required Elements

- QueryFilter

MaterialAddRq

Note: This element requires a certain required element, which can be followed by an optional element, then requires more required elements, which can be followed by additional optional elements in sequence.

Required Elements

- MaterialAdd

Optional Elements

- SetStockItem — Used for **raw stock** only. If not defined, a default Stock Item will be created.

Required Elements

- Choice of:
  - SetShape — Used for **raw stock** only. Default value is Round. For JobBOSS 9.0 and later, we recommend that you maintain this element in StockItemAddRq.
  - AssignRole — Used for **discrete** materials only. Default value is'F'.

Optional Elements

- AssignUnits
- EditDimension — Can have multiple. Used for **raw stock** only.
- EditAlloy — Used for **raw stock** only. For JobBOSS 9.0 and later, we recommend that

you maintain this element in StockItemAddRq.

- SetRevInformation — Used for **finished goods** only.
- AddOnHandQty — Can have multiple.
- WeightFactorMod — Used for **raw stock** only.
- CustomerPartAdd — Can have multiple.
- PriceBreakAdd — Can have multiple. Used for **discrete** materials.
- PurchaseUnitConvMod
- VendorPartAdd — Can have multiple.

MaterialModRq

Note: This element requires a certain required element, which can be followed by an optional element, then requires more required elements, which can be followed by additional optional elements in sequence.

Required Elements

- MaterialMod

Optional Elements

- SetStockItem — Used for **raw stock** only.

Required Elements

- Choice of:
  - SetShape — Used for **raw stock** only.
  - AssignRole

Optional Elements

- AssignUnits
- EditAlloy — Used for **raw stock** only.
- EditDimension — Can have multiple. Used for **raw stock** only.
- SetRevInformation — Used for **finished goods** only.
- AdjustOnHandQty — Can have multiple.
- WeightFactorMod — Used for **raw stock** only.
- CustomerPartMod — Can have multiple.
- CustomerPartDelete — Can have multiple.
- CustomerPartAdd — Can have multiple.
- PriceBreakMod — Can have multiple. Used for **discrete** materials.
- PriceBreakDel — Can have multiple. Used for **discrete** materials.
- PriceBreakAdd — Can have multiple. Used for **discrete** materials.
- PurchaseUnitConvMod

MaterialDelRq

Required Elements

- MaterialDel

MaterialQueryRq

Required Elements

- MaterialQueryFilter

MateriallistQueryRq

Required Elements

- QueryFilter

JobAddRq

For assembly structures, you need to add a job via JobAdd for each component in the assembly, as well as for the top level job. Then additionally you need to add a component via ComponentAdd for each component job to tie it to its parent job.

If you want to add components for an assembly job *at the time of creating the assembly,* you will need to first add all the components as jobs, adding the lowest level components first, and then create the job which uses these components. (Another way to achieve the same result is to add all jobs first using JobAddRq, then use JobModRq to add components where appropriate.) For example, say you have the following component job structure:

Job Number	*Level*
9800	0
9800-A	1
9800-B	1
9800-C	1
9800-C-l	2
9800-C-2	2

To add components 9800-A, 9800-B, and 9800-C to 9800 at the same time as creating 9800, you would need to add 9800-A, 9800-B, and 9800-C as jobs first. Similarly, to add components 9800-C-1 and 9800-C-2 at the same time as creating 9800-C, you would need to add them as jobs first. Thus, the overall order ofrequests would be as follows (lines are used to separate each functional request):

```
<JobAddRq>
  <JobAdd>
    <ID>9800-C-l</ID>
  </JobAdd>
</JobAddRq>
```

```
<JobAddRq>
  <JobAdd>
    <ID>9800-C-2</ID>
  </JobAdd>
</JobAddRq>
```

```
<JobAddRq>
  <JobAdd>
    <ID>9800-A</ID>
  </JobAdd>
</JobAddRq>
```

```
<JobAddRq>
  <JobAdd>
    <ID>9800-B</ID>
  </JobAdd>
```

```
</JobAddRq>
```

```
<JobAddRq>
  <JobAdd>
    <ID>9800-C</ID>
  </JobAdd>
  <ComponentAdd>
    <ComponentJobRef ID="9800-C-1" />
  </ComponentAdd>
  <ComponentAdd>
    <ComponentJobRef ID="9800-C-2" />
  </ComponentAdd>
</JobAddRq>
```

```
<JobAddRq>
  <JobAdd>
    <ID>9800</ID>
  </JobAdd>
  <ComponentAdd>
    <ComponentJobRef ID="9800-A" />
  </ComponentAdd>
  <ComponentAdd>
    <ComponentJobRef ID="9800-B" />
  </ComponentAdd>
  <ComponentAdd>
    <ComponentJobRef ID="9800-C" />
  </ComponentAdd>
</JobAddRq>
```

Required Elements

- **JobAdd**

Optional Elements

- **SetUpCustomerInfo**
- **PartSpecification**
- **BuildQuantities**
- **SetPriceInfo**
- **SetSalesRep**
- **AdditionalChargeAdd** — **Can have multiple.**
- **DeliveryAdd** — **Can have multiple.**
- **RouteAddFrom WorkCenter** — **Can have multiple.**
- **RouteAddFromOperation** — **Can have multiple.**
- **RouteAddFromService** — **Can have multiple.**
- **InsertRouteFromWorkCenter** — **Can have multiple.**
- **InsertRouteFromOperation** — **Can have multiple .**
- **InsertRouteFromService** — **Can have multiple.**
- **MaterialRequirementAdd** — **Can have multiple.**
- **MiscMaterialRequirementAdd** — **Can have multiple.**
- **ComponentAdd** — **Can have multiple.**
- **UsePartMaster** — **Type: boolean.**

JobModRq

Required Elements

- JobMod

Optional Elements

- SetUpCustomerInfo
- PartSpecification
- BuildQuantities
- SetPriceInfo
- SetSalesRep
- AdditionalChargeAdd Can have multiple.
- AdditionalChargeMod Can have multiple.
- AdditionalChargeDel Can have multiple.
- Delivery Add Can have multiple.
- Delivery Mod Can have multiple.
- Delivery Del Can have multiple.
- RouteAddFrom WorkCenter Can have multiple.
- RouteAddFromOperation Can have multiple.
- RouteAddFromService Can have multiple.
- InsertRouteFromWorkCenter Can have multiple.
- InsertRouteFmmOperation Can have multiple.
- InsertRouteFromService Can have multiple.
- InsideRouteMod Can have multiple.
- OutsideRouteMod Can have multiple.
- RouteDel Can have multiple.
- MaterialRequirementAdd Can have multiple.
- MiscMaterialRequirementAdd Can have multiple.
- MaterialRequirementMod Can have multiple.
- MaterialRequirementDel Can have multiple.
- ComponentAdd Can have multiple.
- ComponentMod Can have multiple.
- ComponentDel Can have multiple.
- LinkReq Can have multiple.
- RemoveLink Can have multiple.
- SetUpEntry Add Can have multiple.
- RunTimeEntry Add Can have multiple.
- TimeEntryMod Can have multiple.
- TimeEntry Delete Can have multiple.

JobDelRq

Required Elements

- JobDel

JobQueryRq

Required Elements

- JobQueryFilter

JoblistQueryRq

Required Elements

- QueryFilter

LocationListQueryRq

Required Elements

- SimpleQueryFilter

DepartmentReflistQueryRq

Required Elements

- SimpleQueryFilter

CustomerTypelistQueryRq

Required Elements

- SimpleQueryFilter

TaxCodelistQueryRq

Required Elements

- SimpleQueryFilter

ShipVialistQueryRq

Required Elements

- SimpleQueryFilter

AlloylistQueryRq

Required Elements

- SimpleQueryFilter

MaterialClasslistQueryRq

Required Elements

- SimpleQueryFilter

DiscountlevellistQueryRq

Required Elements

- SimpleQueryFilter

TermslistQueryRq

Required Elements

- SimpleQueryFilter

CurrencylistQueryRq

Required Elements

- SimpleQueryFilter

SalesCodelistQueryRq

Required Elements

- SimpleQueryFilter

LocationTypelistQueryRq

Required Elements

- SimpleQueryFilter

WarehouselistQueryRq

Required Elements

- SimpleQueryFilter

AdditionalCostlistQueryRq

Required Elements

- SimpleQueryFilter

StandardTextlistQueryRq

This query provides a list of standard text entries which you can then use to copy and paste information into your XML request file where needed.

Required Elements

- SimpleQueryFilter

MaterialConversionQueryRq

Required Elements

- MaterialConversionQuery

TimeEntrylistQueryRq

Required Elements

- TimeEntryQueryFilter

SalesOrderAddRq

Required Elements

- SalesOrderAdd
- SetUpCustomerInfo

Optional Elements

- AddSOLine — Can have multiple.
- SOAdditionalChargeAdd — Can have multiple.

SalesOrderModRq

Required Elements

- SalesOrderMod

Optional Elements

- AddSOLine — Can have multiple.
- SOLineMod — Can have multiple.
- SOLineDel — Can have multiple.
- SOAdditionalChargeAdd — Can have multiple.
- AdditionalChargeMod — Can have multiple.
- AdditionalChargeDel — Can have multiple.

SalesOrderDelRq

Required Elements

- SalesOrderDel

SalesOrderQueryRq

Required Elements

- SalesOrderQueryFilter

SalesOrderlistQueryRq

Required Elements

- SOQueryFilter

CustomerAdd

Required Elements

- ID — Type: string.

Optional Elements

- Name Type: string.
- PartnerSince Type: date.
- LeadDays Type: int.
- URL Type: string.
- EMailAddress Type: string.
- Status Type: string.

Status
active
inactive
prospect
warning
pilot
credit hold
reference

- CreditLimit Type: double.
- AcceptsBackorders Type: boolean.
- PrintStatement Type: boolean.
- SalesRepRef
- DiscountLevelRef
- TradingCurrency Ref
- TermsCodeRef
- TaxCodeRef
- CustomerTypeRef
- ShipViaRef
- UserValues

AddressAdd

Business rules for specifying customer and vendor addresses are as follows:

- You can designate an address as billable, shippable, or both.
- An address can act as one or more kind of default address (main, billing, and shipping).
- You need to at least specify a default main address and a default billing address in your system. Specifying a default shipping address is optional.

Note: If after setting a default address in AddressAdd you later use SetAddressDefault, the more recent value (from SetAddressDefault) is the one that will be used.

Required Elements

- AddressName Type: string.

Optional Elements

- Name Type: string.
- Linel Type: string.
- Line2 Type: string.
- City Type: string.
- State Type: string.
- PostalCode Type: string.

- Country Type: string.
- Phone Type: string.
- Fax Type: string.
- Shippable Type: boolean.
- Billable Type: boolean.
- DefaultMain Type: boolean.
- DefaultAccounting Type: boolean.
- DefaultShipping Type: boolean.

ContactAdd

Required Elements

- Name Type: string.

Optional Elements

- Title Type: string.
- PhoneNumber Type: string.
- PhoneExtension Type: string.
- Fax Type: string.
- EMailAddress Type: string.
- AddressRef

StartUpDocumentAdd

Required Elements

- ID Type: string.
- OriginalAmount Type: double.
- OpenAmount Type: double.

Optional Elements

- DocumentDate Type: date.
- DueDate Type: date.
- Description Type: string.
- DocumentType Type: string.

CustomerMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Name Type: string.
- PartnerSince Type: date.
- LeadDays Type: int.
- URL Type: string.
- EMailAddress Type: string.

- Status — Type: string.

Status
active
inactive
prospect
warning
credit hold
reference
pilot

- CreditLimit — Type: double.
- AcceptsBackorders — Type: boolean.
- PrintStatement — Type: boolean.
- SalesRepRef
- DiscountLevelRef
- TradingCurrencyRef
- TermsCodeRef
- TaxCodeRef
- CustomerTypeRef
- ShipViaRef
- UserValues

AddressDel

Required Elements

- ID — Type: string.

AddressMod

Refer to AddressAdd for business rules for specifying customer and vendor addresses.

Required Elements

- ID — Type: string.
- LastUpdated — Type: dateTime.

Optional Elements

- Name — Type: string.
- Line! — Type: string.
- Line2 — Type: string.
- City — Type: string.
- State — Type: string.
- PostalCode — Type: string.
- Country — Type: string.
- Phone — Type: string.
- Fax — Type: string.
- Shippable — Type: boolean.
- Billable — Type: boolean.
- DefaultMain — Type: boolean.
- DefaultAccounting — Type: boolean.

- DefaultShipping Type: boolean.

ContactDel

Required Elements

- ID Type: string.

ContactMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Name Type: string.
- Title Type: string.
- PhoneNumber Type: string.
- PhoneExtension Type: string.
- Fax Type: string.
- EMailAddress Type: string.
- AddressRef

StartUpDocumentDel

Required Elements

- ID Type: string.

StartUpDocumentMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- DueDate Type: date.
- Description Type: string.

CustomerDel

Required Elements

- ID Type: string.

CustomerQueryFilter

Required Elements

- ID Type: string.
- IncludeAddressDetails Type: boolean.

- IncludeContacts Type: boolean.
- IncludeStartUpDocuments Type: boolean.

QueryFilter

Required Elements

- IDFilter Type: string.
- NameFilter Type: string.
- TypeFilter Type: string.

VendorAdd

Note: If you plan to add a service contract for a new vendor, you need to add the service contract first, and then add the vendor that uses the service contract.

Required Elements

- ID Type: string.

Optional Elements

- Name Type: string.
- PartnerSince Type: date.
- LeadDays Type: int.
- URL Type: string.
- EMailAddress Type: string.
- Status Type: string.

Value
Active
Inactive

- Send1099 Type: boolean.
- AccountNbr Type: string.
- TaxIDNbr Type: string.
- GLAccountNbr Type: string.
- TermsCodeRef
- ShipViaRef
- TaxCodeRef
- TradingCurrencyRef
- UserValues

ServiceContractAdd

Required Elements

- ServiceRef

Optional Elements

- MinimumCharge Type: double.
- LeadDays Type: int.

VendorStartU pDocAdd

Note: This element requires certain required elements, which can be followed by some optional elements, then requires another required element, which can be followed by additional optional elements in sequence.

Required Elements

- DocumentNbr Type: string.
- OriginalAmount Type: double.

Optional Elements

- OpenAmount Type: double.
- DiscountAmount Type: double.

Required Elements

- DocumentDate Type: date.

Optional Elements

- DueDate Type: date.
- Description Type: string.
- DocumentType Type: string.
- DiscountDueDate Type: date.

VendorMod

Note: If you plan to add a service contract for an existing vendor, you need to add the service contract first, and then modify the vendor that uses the service contract.

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Name Type: string.
- PartnerSince Type: date.
- LeadDays Type: int.
- URL Type: string.
- EMailAddress Type: string.
- Status Type: string.

Value
Active
Inactive

- Send1099 Type: boolean.
- AccountNbr Type: string.
- TaxIDNbr Type: string.
- GLAccountNbr Type: string.
- TermsCodeRef

- ShipViaRef
- TaxCodeRef
- TradingCurrencyRef
- UserValues

VendorStartUpDocMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- DueDate Type: date.

ServiceContractDel

Required Elements

- ID Type: string.

SeviceContractMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- MinimumCharge Type: double.
- LeadDays Type: int.

VendorDel

Required Elements

- ID Type: string.

VendorQueryFilter

Required Elements

- ID Type: string.
- IncludeAddressDetails Type: boolean.
- IncludeContacts Type: boolean.
- IncludeStartUpDocuments Type: boolean.

Optional Elements

- IncludeVendorParts Type: boolean.

ServiceOperationAdd

Required Elements

- ID Type: string.
- Description Type: string.

Optional Elements

- LeadDays Type: int.
- DefaultMinCharge Type: double.
- Notes Type: string.
- UserValues

ServiceOperationMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Description Type: string.
- LeadDays Type: int.
- DefaultMinCharge Type: double.
- Notes Type: string.
- UserValues

ServiceOperationDel

Required Elements

- ID Type: string.

ServiceOperationQueryFilter

Required Elements

- ID Type: string.
- IncludeContracts Type: boolean.

WorkCenterAdd

Required Elements

- ID Type: string.

Optional Elements

- LinkMaterial Type: boolean.
- LinkComponents Type: boolean.
- Type Type: string.

Value
Direct

Indirect

- Notes Type: string.
- DepartmentRef
- Capacity
- BurdenCosts
- LaborCosts
- UserValues

WorkCenterMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- LinkMaterial Type: boolean.
- LinkComponents Type: boolean.
- Type Type: string.

Value
Direct
Indirect

- Notes Type: string.
- DepartmentRef
- Capacity
- BurdenCosts
- LaborCosts
- UserValues

WorkCenterDel

Required Elements

- ID Type: string.

WorkCenterQueryFilter

Required Elements

- ID Type: string.

StockItemAdd

Required Elements

- Name Type: string.
- DefaultLocationRef
- MaterialClassRef

Optional Elements

- DefaultSource Type: string.

- LotControlled Type: boolean.
- RoundtoWholeUnit Type: boolean.
- LeadDays Type: int.
- GLAccountID Type: string.
- StockUnit Type: string.
- PurchaseUnit Type: string.
- CostUnit Type: string.
- VendorRef
- SalesCodeRef

StockItemMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Name Type: string.
- DefaultLocationRef
- MaterialClassRef
- DefaultSource Type: string.
- LotControlled Type: boolean.
- RoundtoWholeUnit Type: boolean.
- LeadDays Type: int.
- GLAcccountID Type: string.
- StockUnit Type: string.
- PurchaseUnit Type: string.
- CostUnit Type: string.
- VendorRef
- SalesCodeRef

StockItemDel

Required Elements

- ID Type: string.

StockItemQueryFilter

Required Elements

- ID Type: string.
- IncludeMaterialItems Type: boolean.
- IncludeVendorParts Type: boolean.

MaterialAdd

Note: This element requires certain required elements, which can be followed by some optional elements, then requires another required element, which can be followed by

additional optional elements in sequence.

Required Elements

- ID — Type: string.
- MaterialType — Type: string.

Value	*Description*
F	finished goods
H	hardware
S	supplies
R	raw stock

Optional Elements

- Description — Type: string.
- ExtendedDescription — Type: string.
- VendorReferenceNbr — Type: string.
- DefaultSource — Type: string.
- LotControlled — Type: boolean.
- Status — Type: string.

Value
Active
Inactive

- RoundtoWholeUnit — Type: boolean.
- StandardCost — Type: double.
- AverageCost — Type: double.
- LastCost — Type: double.
- SellPrice — Type: double.
- LeadDays — Type: int.
- Notes — Type: string.
- GLAccountID — Type: string.
- VendorRef

Required Elements

- DefaultLocationRef

Optional Elements

- MaterialClassRef — For JobBOSS 9.0 and later, we recommend that you maintain this element in StockitemAddRq.
- SalesCodeRef
- ReorderPoint — Type: double.
- ReorderQty — Type: double.
- UserValues
- PartMasterRef

SetStockItem

Required Elements

- ID Type: string. References a list of materials.

SetShape

Required Elements

- ShapeName Type: string.

AssignRole

Required Elements

- RoleType Type: string.

Value	*Description*
F	finished goods
H	hardware
S	supplies

AssignUnits

Optional Elements

- StockUnit Type: string.
- PurchaseUnit Type: string.
- CostUnit Type: string.
- PriceUnit Type: string.
- PriceUnitConversion Type: double.

EditDimension

Required Elements

- DimensionName Type: string.

This table lists each shape name with the dimension names associated with it:

Shape Name	**Dimension Names**			
Angle	Thickness	Bar Length	Leg 1	Leg2
Channel	Flange	Bar Length	Web Thickness	Depth
Coil	Width	Thickness		
Extrusion	Bar Length			
HBeam	Flange	Bar Length	Thickness	Web
Hexagon	Flat-to-flat	Bar Length		
I Beam	Flange	Bar Length	Thickness	Web
Octagon	Flat-to-flat	Bar Length		
Preform	Width	Length	Height	
Rec Tube	Height	Bar Length	Wall	Width
Rectangle	Width	Bar Length	Thickness	
Round	Diameter	Bar Length		
Sheet	Width	Length	Thickness	
Sq Tubing	Outside	Bar Length	Inside	
TBeam	Flange	Bar Length	Thickness	Web

Triangle	Side A	Bar Length	Angle Between	Side B
Tubing	OD	Bar Length	Wall	
Wire	Diameter			
ZBeam	Height	Bar Length	Thickness	Flange

- Value Type: double.
- UnitofMeasure Type: string.

Value	*Description*
in	inch
cm	centimeter

EditAlloy

Required Elements

- AlloylD Type: string.

SetRevInformation

Optional Elements

- Revision Type: string.
- Drawing Type: string.

AddOnHandQty

Required Elements

- LocationRef
- Quantity Type: double.

Optional Elements

- LotNumber Type: string.

WeightFactorMod

Required Elements

- Value Type: double.

VendorPartAdd

Required Elements

- VendorRef

Optional Elements

- VendorReferenceNbr Type: string.
- LeadDays Type: int.
- PriceUnit Type: string.
- SellPrice Type: double.

VendorPartMod

Note: You cannot use VendorPartMod to change the vendor ID for a vendor part. Instead, use VendorPartDelete to delete the vendor part and then use VendorPartAdd to add it again with the correct vendor ID.

Required Elements

- ID Type: string. Internal ID for vendor part.
- LastUpdated Type: dateTime.

Optional Elements

- VendorReferenceNbr Type: string.
- LeadDays Type: int.
- PriceUnit Type: string.
- SellPrice Type: double.

VendorPartDelete

Required Elements

- ID Type: string.

CustomerPartAdd

Required Elements

- Choice of:
  - CustomerRef
  - CustomerTypeRef
- TheirPartNumber Type: string.

Optional Elements

- TheirDescription Type: string.
- TheirExtDescription Type: string.
- TheirRevision Type: string.
- TheirPrice Type: double.

PriceBreakAdd

Required Elements

- MinimumQuantity Type: double.
- Price Type: double.

PurchaseUnitConvMod

Required Elements

- Value Type: double.

MaterialMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Description Type: string.
- ExtendedDescription Type: string.
- VendorReferenceNbr Type: string.
- DefaultSource Type: string.
- LotControlled Type: boolean.
- Status Type: string.

Value
Active
Inactive

- RoundtoWholeUnit Type: boolean.
- StandardCost Type: double.
- AverageCost Type: double.
- LastCost Type: double.
- SellPrice Type: double.
- LeadDays Type: int.
- Notes Type: string.
- GLAccountID Type: string.
- VendorRef
- DefaultLocationRef
- MaterialClassRef
- SalesCodeRef
- ReorderPoint Type: double.
- ReorderQty Type: double.
- UserValues
- PartMasterRef

AdjustOnHandQty

Required Elements

- ReasonRef
- Quantity Type: double.

Optional Elements

- MaterialLocationRef
- TransactionDate Type: date.

CustomerPartMod

Required Elements

- ID Type: string.

- LastUpdated Type: dateTime.

Optional Elements

- TheirPartNumber Type: string.
- TheirDescription Type: string.
- TheirExtDescription Type: string.
- TheirRevision Type: string.
- TheirPrice Type: double.

CustomerPartDelete

Required Elements

- ID Type: string.

PriceBreakMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- MinimumQuantity Type: double.
- Price Type: double.

PriceBreakDel

Required Elements

- ID Type: string.

MaterialDel

Required Elements

- ID Type: string.

MaterialQueryFilter

Required Elements

- ID Type: string.
- IncludeMaterialLocations Type: boolean.
- IncludeCustomerParts Type: boolean.
- IncludePriceBreaks Type: boolean.

JobAdd

To have JobBOSS generate automatically numbered IDs for you (instead of your having to explicitly specify them), enter a blank character as the value for the ID. For example:

```
.:: <JobAddRq>
.:: <JobAdd>
   <ID> </ID>
```

```
(Optional elements for JobAdd go here..).
  </JobAdd>
(Optional elements for JobAddRq go here...)
  </JobAddRq>
```

Required Elements

- ID Type: string.

Optional Elements

- BuildtoStock Type: boolean.
- CustomerPO Type: string.
- CustomerPOLine Type: string.
- TimeandMaterial Type: boolean.
- Quote Type: string.
- Status Type: string.

Value
Pending
Active
Hold
Complete
Closed
Template

- Comments Type: string.
- Priority Type: int.
- CertsReqd Type: boolean.
- Notes Type: string.
- LeadDays Type: int.
- OrderDate Type: date.
- OrderQty Type: int.
- ShipViaRef
- PricingCalculator
- UserValues
- Source Type: string.

Value	*Description*
SDK	Specifies the data source is XML formatted data added using the SDK.

SetUpCustomerInfo

Required Elements

- CustomerRef
- OverrideCreditLimit Type: boolean.

Optional Elements

- CountactRef
- ShipToRef
- ShipViaRef
- SalesRepRef
- TermsCodeRef

- TaxCodeRef
- SalesCodeRef
- ExchangeContract

PartSpecification

Required Elements

- PartNumber Type: string.

Optional Elements

- Description Type: string.
- ExtendedDescription Type: string.
- Revision Type: string.
- DrawingNbr Type: string.
- ScrapPercent Type: string.

BuildQuantities

Optional Elements

- ExtraQuantity Type: int.
- PickQty Type: int.

SetPriceInfo

Optional Elements

- OrderUnit Type: string.
- UnitPrice Type: double.
- PriceUofM Type: string.
- ConversionRatio Type: double.

SetSalesRep

Optional Elements

- SalesRepRef
- Commission Type: string.

AdditionalChargeAdd

Optional Elements

- Amount Type: double.
- Description Type: string. Valid values are user defined.
- Taxable Type: boolean.
- Commissionable Type: boolean.
- JobRevenue Type: boolean.
- Notes Type: string.
- SalesCodeRef

DeliveryAdd

Optional Elements

- RequestedDate Type: date.
- PromisedDate Type: date.
- PromisedQuantity Type: int.

RouteAddFromWorkCenter

Required Elements

- WorkCenterRef

Optional Elements

- InsideRouteProperties

RouteAddFromOperation

Required Elements

- OperationRef

Optional Elements

- InsideRouteProperties

RouteAddFromService

Required Elements

- ServiceRef
- VendorRef

Optional Elements

- OutsideRouteProperties

InsertRouteFromWorkCenter

Required Elements

- WorkCenterRef
- Sequence Type: int.

Optional Elements

- InsideRouteProperties

InsertRouteFromOperation

Required Elements

- OperationRef
- Sequence Type: int.

Optional Elements

- InsideRouteProperties

InsertRouteFromService

Required Elements

- ServiceRef
- VendorRef
- Sequence Type: int.

Optional Elements

- OutsideRouteProperties

MaterialRequirementAdd

Required Elements

- MaterialRef

Optional Elements

- DefaultSource Type: string.
- RequirementProperties

MiscMaterialRequirementAdd

Required Elements

- MaterialRef

Optional Elements

- Description Type: string.
- RequirementProperties

ComponentAdd

Required Elements

- ComponentJobRef

Optional Elements

- QuantityPer Type: int.

JobMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- BuildtoStock Type: boolean.

- CustomerPO Type: string.
- CustomerPOLine Type: string.
- TimeandMaterial Type: boolean.
- Quote Type: string.
- Status Type: string.
- Comments Type: string.
- Priority Type: int.
- CertsReqd Type: boolean.
- Notes Type: string.
- LeadDays Type: int.
- OrderDate Type: date.
- OrderQty Type: int.
- ShipViaRef
- PricingCalculator
- UserValues

AdditionalChargeMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Amount Type: double.
- Description Type: string.
- Taxable Type: boolean.
- Commissionable Type: boolean.
- JobRevenue Type: boolean.
- Notes Type: string.
- SalesCodeRef

AdditionalChargeDel

Required Elements

- ID Type: string.

DeliveryMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- RequestedDate Type: date.
- PromisedDate Type: date.
- Promised Quantity Type: int.

DeliveryDel

Required Elements

- ID Type: string.

InsideRouteMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- InsideRouteProperties

OutsideRouteMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- OutsideRouteProperties

RouteDel

Required Elements

- ID Type: string.

MaterialRequirementMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- DefaultSource Type: string.
- Description Type: string.
- RequirementProperties

MaterialRequirementDel

Required Elements

- ID Type: string.

ComponentMod

Required Elements

- ID Type: string.

- LastUpdated Type: dateTime.
- QuantityPer Type: int.

ComponentDel

Required Elements

- ID Type: string.

LinkReq

Required Elements

- OperationRef
- RequirementRef
- Link.Type Type: string.

Value
MaterialReq
Job

Removelink

Required Elements

- OperationRef
- RequirementRef
- Link.Type Type: string.

Value
MaterialReq
Job

SetUpEntryAdd

Required Elements

- OperationRef
- EmployeeRef

Optional Elements

- SetUpTimeEntryProperties

RunTimeEntryAdd

Required Elements

- OperationRef
- EmployeeRef

Optional Elements

- RunTimeEntryProperties

TimeEntryMod

Required Elements

- OperationRef
- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- RunTimeEntryProperties
- SetUpTimeEntryProperties

TimeEntryDelete

Required Elements

- OperationRef
- ID Type: string.

JobDel

Required Elements

- ID Type: string.

JobQueryFilter

Required Elements

- ID Type: string.
- IncludeAdditionalCharges Type: boolean.
- IncludeDeliveries Type: boolean.
- IncludeRoutingLines Type: boolean.
- IncludeMaterialRequirements Type: boolean.
- IncludeComponents Type: boolean.

SimpleQueryFilter

Optional Elements

- IDFilter Type: string.

SalesRepRef

Required Attributes

- Name= "ID" Type: string.

DiscountlevelRef

Required Attributes

- Name= "ID" Type: string. Valid values are user defined.

TradingCurrencyRef

Required Attributes

- Name = "ID" Type: string. Valid values are user defined.

TermsCodeRef

Required Attributes

- Name= "ID" Type: string. Valid values are user defined.

TaxCodeRef

Required Attributes

- Name = "ID" Type: string. Valid values are user defined.

CustomerTypeRef

Required Attributes

- Name = "ID" Type: string. Valid values are user defined.

ShipViaRef

Required Attributes

- Name= "ID" Type: string. Valid values are user defined.

UserValues

Optional Elements

- Datel Type: date.
- Date2 Type: date.
- Textl Type: string.
- Text2 Type: string.
- Text3 Type: string.
- Text4 Type: string.
- Text5 Type: string.
- Amountl Type: double.
- Amount2 Type: double.
- Numericl Type: double.
- Numeric2 Type: double.
- Decimall Type: double.
- Notes Type: string.

AddressRef

Required Attributes

- Name= "ID" Type: string.

ServiceRef

Required Attributes

- Name = "ID" Type: string.

DepartmentRef

Required Attributes

- Name= "ID" Type: string.

Capacity

Optional Elements

- LimitingResources* Type: int.
- SchedulingHours* Type: int.
- QueuHours Type: int.

*As of JobBOSS 11.0, the field is no longer used. However, the field does not need to be removed from your application.

BurdenCosts

Optional Elements

- LaborBurdenRate Type: double.
- MachineBurdenRate Type: double.
- GABurdenR.ate Type: double.

LaborCosts

Optional Elements

- SetUpLaborRate Type: double.
- RunLaborRate Type: double.

VendorRef

Required Attributes

- Name = "ID" Type: string.

DefaultlocationRef

Required Attributes

- Name = "ID" Type: string.

MaterialClassRef

Required Attributes

- Name= "ID" Type: string. Valid values are user defined.

SalesCodeRef

Required Attributes

- Name = "ID" Type: string. Valid values are user defined.

PartMasterRef

Required Attributes

- Name= "ID" Type: string. Valid ID of a job whose status is template.

LocationRef

Required Attributes

- Name= "ID" Type: string.

CustomerRef

Required Attributes

- Name= "ID" Type: string.

ReasonRef

Required Attributes

- Name= "ID" Type: string. Valid values are user defined.

MaterialLocationRef

Required Attributes

- Name = "ID" Type: string.

PricingCalculator

Optional Elements

- ProfitorMarkup Type: string.
- ProfitPercent Type: double.
- LaborMarkup Type: double.
- MaterialMarkup Type: double.
- ServiceMarkup Type: double.
- LaborBurdenMarkup Type: double.
- MachineBurdenMarkup Type: double.
- GABurdenMarkup Type: double.

CountactRef

Required Attributes

- Name= "ID" Type: string.

ShipToRef

Required Attributes

- Name = "ID" Type: string.

ExchangeContract

Optional Elements

- TradeCurrencyRef
- Fixed Type: boolean.
- Rate Type: double.
- TradeDate Type: date.

OutsideRouteSchedule

Required Elements

- StartDate Type: date.
- EndDate Type: date.

Optional Elements

- StartTime Type: double.
- EndTime Type: double.

WorkCenterRef

Required Attributes

- Name= "ID" Type: string.

InsideRouteProperties

Optional Elements

- Operation Type: string.
- Description Type: string.
- Status Type: string.

Value	*Description*
0	open
S	started
C	closed

- Priority Type: int.
- PercentAttended Type: int.
- Resources to Use Type: double.
- SetUpCalculator
- RunTimeCalculator
- OverlapSpecifier
- RouteSchedule
- Notes Type: string.
- FloorNotes Type: string.

Operation Ref

Required Attributes

- Name = "ID" Type: string.

RequirementRef

Required Attributes

- Name= "ID" Type: string.

EmployeeRef

Required Attributes

- Name = "ID" Type: string.

SetUpTimeEntryProperties

Optional Elements

- WorkDate Type: dateTime.
- Hours Type: double.
- LaborHours Type: double.
- OTFactor Type: double.
- OTHours Type: double.
- Rework Type: boolean.
- ReworkReasonRef
- PercentComplete Type: double.
- Quantity Type: int.
- ScrapQuantity Type: int.
- ScrapReasonRef

Run TimeEntryProperties

Optional Elements

- WorkDate Type: date.
- Hours Type: double.
- LaborHours Type: double.
- OTFactor Type: double.
- OTHours Type: double.
- Rework Type: boolean.
- ReworkReasonRef
- PercentComplete Type: double.
- Quantity Type: int.
- ScrapQuantity Type: int.
- ScrapReasonRef
- Complete Type: boolean.
- AttendedPercent Type: double.

OutsideRouteProperties

Optional Elements

- Description — Type: string.
- Status — Type: string.

Value	*Description*
0	open
S	started
C	closed

- UnitCost — Type: double.
- CostUnit — Type: string.
- PurchaseUnit — Type: string.
- ConversionFactor — Type: double.
- LeadDays — Type: int.
- CertsReqd — Type: boolean.
- Notes — Type: string.
- MinimumCharge — Type: double.
- VendorRef
- AdditionalCost
- ExchangeContract
- OutsideRouteSchedule
- FloorNotes — Type: string.

ReworkReasonRef

Required Attributes

- Name = "ID" — Type: string. Valid values are user defined.

ScrapReasonRef

Required Attributes

- Name= "ID" — Type: string. Valid values are user defined.

MaterialRef

Required Attributes

- Name = "ID" — Type: string.

ShipToAddress

Optional Attributes

- Name = "ID" — Type: string.

RequirementProperties

Optional Elements

- VendorReference Type: string.
- CertsReqd Type: boolean.
- Quantity Type: double.
- UnitofMeasure Type: string. Valid values are user defined.

Note: This element can only be changed for requirement types of miscellaneous.

- UnitPrice Type: double.
- Status Type: string.

Value	*Description*
0	open
S	started
C	closed

- DueDate Type: date.
- LeadDays Type: int.
- VendorRef
- AdditionalCost
- ExchangeContract

ComponentJobRef

Required Attributes

- Name= "ID" Type: string.

TradeCurrencyRef

Required Attributes

- Name = "ID" Type: string.

SetUpCalculator

Required Elements

- Hours Type: float.

RunTimeCalculator

Optional Elements

- RunMethod Type: string.
- Run Type: float.
- Efficiency Type: int.

OverlapSpecifier

Optional Elements

- Method Type: string.
- Overlap Type: float.

RouteSchedu le

Required Elements

- StartDate Type: date.
- EndDate Type: date.

Optional Elements

- StartTime Type: double.
- EndTime Type: double.

AdditionalCost

Required Elements

- Description Type: string.
- Amount Type: double.

MaterialConversionQuery

This element converts a value from one unit of measure to another.

When you specifiy an initial amount and unit, along with a target unit of measure, an amount in the target unit of measure is returned with the target unit of measure.

For example, if you specify a value of **1** foot (amount is **1** and unit is foot), and the target unit of measure is inches, the amount returned would be 12.

Required Elements

- ID Type: string.
- Amount Type: double.
- Unit Type: string.
- TargetUnit Type: string.

TimeEntryQueryFilter

Optional Elements

- IDFilter Type: string.
- JobFilter Type: string.
- OperationF ilter Type: string.
- EmployeeFilter Type: string.
- WorkCenterFilter Type: string.
- BeginDateFilter Type: dateTime.
- EndDateFilter Type: dateTime.

SalesOrderAdd

Required Elements

- ID Type: string.

Optional Elements

- SOHeaderProperties

AddSOLine

Required Elements

- MaterialRef

Optional Elements

- ShipToAddress
- SOLineProperties
- SetQuantityinfo
- ReqSatisfierAssign

SOAdditionalChargeAdd

Required Elements

- LineNumber Type: string.
- AdditionalChargeProperty

SalesOrderMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- SOHeaderProperties

SOLineMod

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- SOLineProperties
- SetQuantitylnfo
- RemoveReqSatisfier Empty tag.
- ReqSatisfierAssign
- PickMaterial

SOLineDel

Required Elements

- ID Type: string.

AdditionalChargeMod

Required Elements

- LineNumber Type: string.
- ID Type: string.
- LastUpdated Type: string.
- AdditionalChargeProperty

AdditionalChargeDel

Required Elements

- ID Type: string.

SalesOrderQueryFilter

Required Elements

- ID Type: string.

Optional Elements

- IncludeDetai!Lines Type: boolean.

SalesOrderDel

Required Elements

- ID Type: string.

SOQueryFilter

Optional Elements

- IDFilter Type: string.
- CustomerFilter Type: string.
- StatusFilter Type: string.

SOHeaderProperties

Optional Elements

- CustomerPO Type: string.
- OrderDate Type: date.
- Status Type: string.

Value
Open
Hold
Closed

- DefaultPromiseDate Type: date.
- OrderTakerRef
- Notes Type: string.
- Comments Type: string.

- UserValues

SOLineProperties

Optional Elements

- LineNumber — Type: string.
- Taxable — Type: boolean.
- Commissionable — Type: boolean.
- Commission — Type: double.
- DropShip — Type: boolean.
- Notes — Type: string.
- CurrentPromisedDate — Type: dateTime.
- Status — Type: string.

Value
Backorder
Shipped
Open
Hold
Closed

- SalesCodeRef

SetQuantityInfo

Optional Elements

- OrderQty — Type: int.
- UnitPrice — Type: double.
- PriceUofM — Type: string.
- PriceUnitConversion — Type: double.
- DiscountPercent — Type: double.

ReqSatisfierAssign

This element links the given job or PO to the specified SOLine.

Optional Elements

- ID — Type: string.
- SatisfierType — Type: string.

Value
Job
PO

AdditionalChargeProperty

Optional Elements

- Amount — Type: double.
- Description — Type: string. Valid values are user defined.
- Taxable — Type: boolean.
- Commissionable — Type: boolean.

- JobRevenue Type: boolean.
- Notes Type: string.
- SalesCodeRef

OrderTakerRef

Note: This is the same as an employee reference, representing an employee.

Required Attributes

- Name= "ID" Type: string

PickMaterial

If you specify a location for this element, the system will pick materials from that location. If not enough materials are present in the specified location, an error is returned.

If you don't specify a location, the system will go through and pick materials from all available locations in order from the one with the highest quantity on hand to the lowest.

Optional Elements

- LocationRef
- Quantity Type: double.
- Lot Type: string.
- PickDate Type: dateTime.

OrderTakerRef

Required Attributes

- Name= "ID" Type: string.

QuoteAddRq

Note: This element requires a certain required element, which can be followed by an optional element, then requires more required elements, which can be followed by an additional optional element in sequence.

Required Elements

- QuoteAdd

Optional Elements

- QuoteSetUpCustomerinfo

Required Elements

- QuoteLineitemAdd Can have multiple.
- QuoteQuantityAdd Can have multiple.

Optional Elements

- QuoteAdditionalChargeAdd Can have multiple.

QuoteModRq

Required Elements

- QuoteMod

Optional Elements

- QuoteSetUpCustomerInfo
- QuoteLineItemAdd Can have multiple.
- QuoteLineItemMod Can have multiple.
- QuoteQuantity Add Can have multiple.
- QuoteQuantityMod Can have multiple.
- QuoteAdditionalChargeAdd Can have multiple.
- QuoteAdditionalChargeMod Can have multiple.

QuoteDelRq

Required Elements

- QuoteDel

QuoteQueryRq

Required Elements

- QuoteQueryFilter

QuotelistQueryRq

Required Elements

- QueryFilter

QuoteAdd

To have JobBOSS generate automatically numbered IDs for you (instead of your having to explicitly specify them), enter a blank character as the value for the ID. For example:

```
=. <QuoteAddRq>
=. <QuoteAdd>
  <ID> </ID>
(Optional elements for QuoteAdd go here...)
  </QuoteAdd>
(Optional elements for QuoteAddRq go here...)
  </QuoteAddRq
```

Required Elements

- ID Type: string.

Optional Elements

- Reference Type: string.
- QuotedBy Type: string.
- DueDate Type: date.
- SubmittedDate Type: date.

- ExpirationDate Type: date.
- RFQDate Type: date.
- QuoteDate Type: date.
- StatusDate Type: date.
- Comments Type: string.
- Notes Type: string.
- LeadDays Type: int.
- CertsReqd Type: boolean.
- Priority Type: int.
- Status Type: string.

Value
Active
Lost
Expired
Hold
Template
Won
Pending
No Quote

- FollowupDate Type: date.
- ActFollowupDate Type: date.
- TermsCodeRef
- ShipToRef
- ShipViaRef
- SalesRepRef
- Commission Type: double.
- SalesCodeRef
- UserValues
- Source Type: string.

Value	*Description*
SDK	Specifies the data source is XML formatted data added using the SDK.

QuoteSetU pCustomerInfo

Required Elements

- CustomerRef
- OverrideCreditLimit Type: boolean.

Optional Elements

- CountactRef
- ShipToRef
- ShipViaRef
- SalesRepRef
- TermsCodeRef
- TaxCodeRef
- SalesCodeRef
- ExchangeContract

QuotelineItemAdd

Required Elements

- QuoteLineItemProperties

QuoteQuantity Add

Required Elements

- QuoteQuantityProperties

QuoteAdditionalChargeAdd

Required Elements

- LineItemID Type: string.

Optional Elements

- Amount Type: double.
- Description Type: string. Valid values are user defined.
- Commissionable Type: boolean.
- JobRevenue Type: boolean.
- Notes Type: string.
- SalesCodeRef

SetAddressDefault

Note: If you first set a default address in AddressAdd, then later use SetAddressDefault, the more recent value (from SetAddressDefault) is the one that will be used.

Required Elements

- ShipToID Type: string.
- DefaultType Type: string.

Value
Main
Ship
Bill

QuotelineItemProperties

Required Elements

- LineitemID Type: string. Specify this temporary IDwhen adding a quote.

Optional Elements

- LastUpdated Type: dateTime.
- LineNumber Type: string.
- PartNumber Type: string.
- QuotedBy Type: string.
- Comments Type: string.

- Notes Type: string.
- CertsReqd Type: boolean.
- LeadDays Type: int.
- Status Type: string.

Value
Won
Lost
Open
Hold
Closed

- SalesCodeRef
- StatusReasonRef
- UserValues
- PartDescription Type: string.
- PartExtDescription Type: string.
- PartDrawingNumber Type: string.
- PartRevision Type: string.
- UsePartMaster Type: boolean.

QuoteQuantityProperties

Required Elements

- LineItemID Type: string.
- Quoted Quantity Type: int.

Optional Elements

- LastUpdated Type: dateTime.
- DecimalPlaces Type: double.
- QuotePrice Type: double.
- OrderUnit Type: string.
- PriceUnit Type: string.
- PriceUnitConversion Type: double.
- Yield Type: double.
- Notes Type: string.

QuoteMod

Required Elements

- QuoteData

QuotelineItemMod

Required Elements

- LineItemID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- LineNumber Type: string.
- PartNumber Type: string.
- QuotedBy Type: string.
- Comments Type: string.
- Notes Type: string.
- CertsReqd Type: boolean.
- LeadDays Type: int.
- Status Type: string.

Value
Won
Lost
Open
Hold
Closed

- SalesCodeRef
- StatusReasonRef
- UserValues
- PartDescription Type: string.
- PartExtDescription Type: string.
- PartDrawingNumber Type: string.
- PartRevision Type: string.

QuoteData

Required Elements

- ID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Reference Type: string.
- QuotedBy Type: string.
- DueDate Type: date.
- SubmittedDate Type: date.
- ExpirationDate Type: date.
- RFQDate Type: date.
- QuoteDate Type: date.
- StatusDate Type: date.
- Comments Type: string.
- Notes Type: string.
- LeadDays Type: int.
- CertsReqd Type: boolean.
- Priority Type: int.
- Status Type: string.

Value
Active
Lost

Expired
Hold
Template
Won
Pending
No Quote

- FollowupDate Type: date.
- ActFollowupDate Type: date.
- TermsCodeRef
- ShipToRef
- ShipViaRef
- SalesRepRef
- Commission Type: double.
- SalesCodeRef
- UserValues
- Source Type: string.

Value	*Description*
SDK	Specifies the data source is XML formatted data added using the SDK.

QuoteQuantityMod

Required Elements

- ID Type: string.
- LineItemID Type: string.
- LastUpdated Type: dateTime.
- QuotedQuantity Type: int.

Optional Elements

- DecimalPlaces Type: double.
- QuotePrice Type: double.
- OrderUnit Type: string.
- PriceUnit Type: string.
- PriceUnitConversion Type: double.
- Yield Type: double.
- Notes Type: string.

QuoteAdditionalChargeMod

Required Elements

- ID Type: string.
- LineItemID Type: string.
- LastUpdated Type: dateTime.

Optional Elements

- Amount Type: double.
- Description Type: string. Valid values are user defined.
- Commissionable Type: boolean.

- JobRevenue Type: boolean.
- Notes Type: string.
- SalesCodeRef

StatusReasonRef

Required Attributes

- Name= "ID" Type: string. Valid values are user defined.

QuoteDel

Required Elements

- ID Type: string.

QuoteQueryFilter

Required Elements

- ID Type: string.
- IncludeLineitems Type: boolean.
- IncludeQuantities Type: boolean.
- IncludeAdditionalCharges Type: boolean.

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